



EE 577B VLSI system Design
Project Phase 1
Team 20

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1.Introduction.

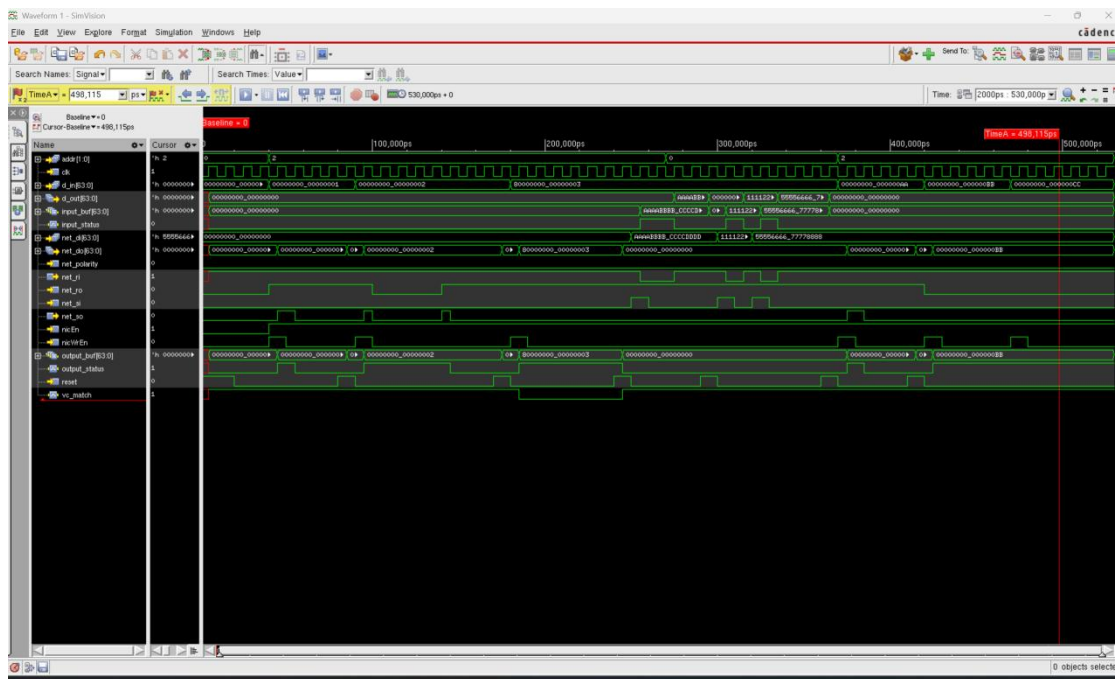
The project phase aims to design and implement a foundational Network-on-Chip (NoC) communication architecture to deepen our understanding of data transmission mechanisms within hardware systems. In this phase, we utilize the cardinal router as the fundamental building block, interconnecting multiple routers to form a 4 times 4 *torus* network topology, thereby realizing a two-dimensional network structure characterized by wraparound connectivity. Concurrently, a Network Interface Controller (NIC) is designed and implemented to facilitate data exchange between the processor and the network.

This report details the aforementioned system, elucidating its overall architectural design, the functionality of key modules, and the implementation methodologies employed. Through the integration and verification of the router, the torus interconnection structure, and the NIC module, this project establishes a practical foundation for comprehending data routing and communication mechanisms within NoC environments, while also laying the groundwork for the design of more complex systems in our future study.

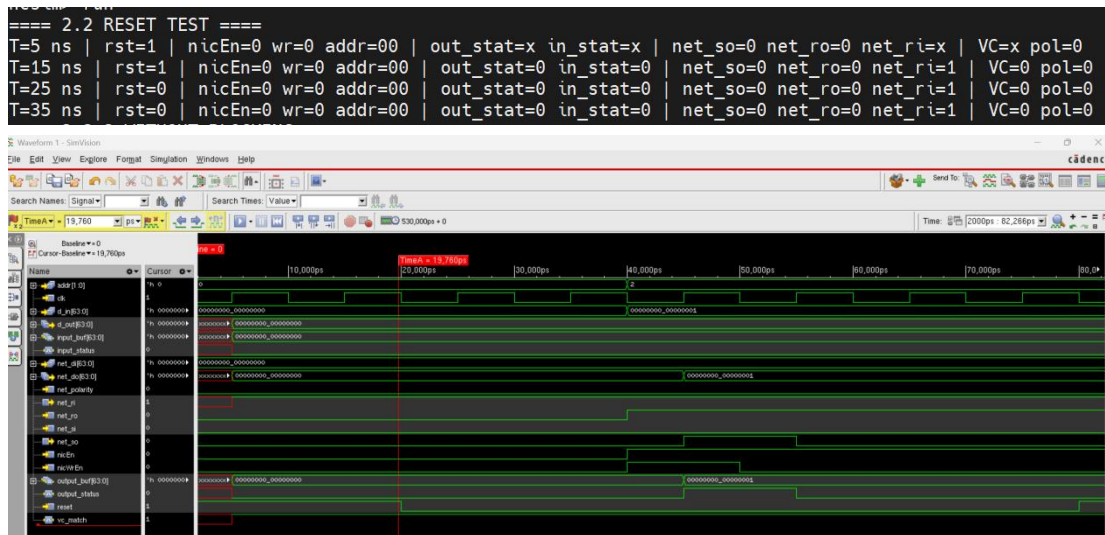
2.Cardinal NIC.

We successfully verified the functionality of the Cardinal NIC, with our implementation passing all required test cases and demonstrating correct behavior under the specified conditions.

2.1.Overall waveform.



2.2.Waveform showing the correct operation of the reset signal.



2.3.Waveform showing the handshake between NIC and router.

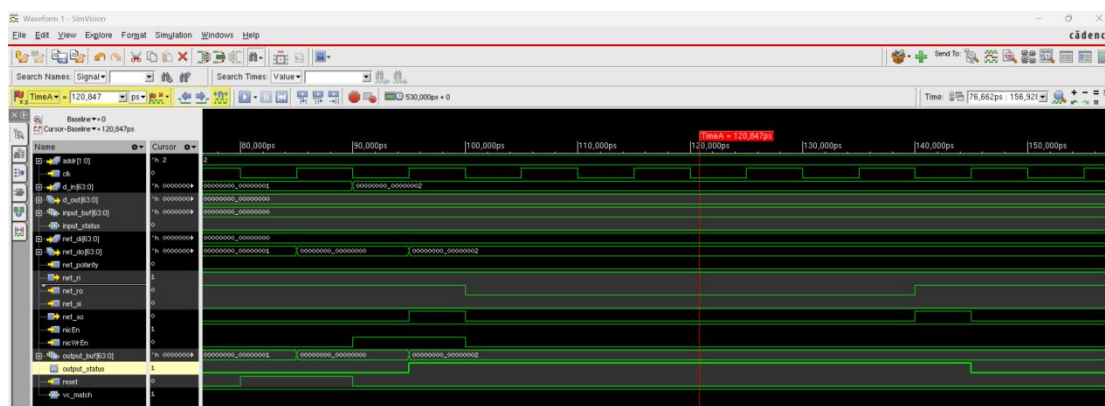
2.3.1.With Blocking.

2.3.1.1.router does not prepared to receive (NIC to router).

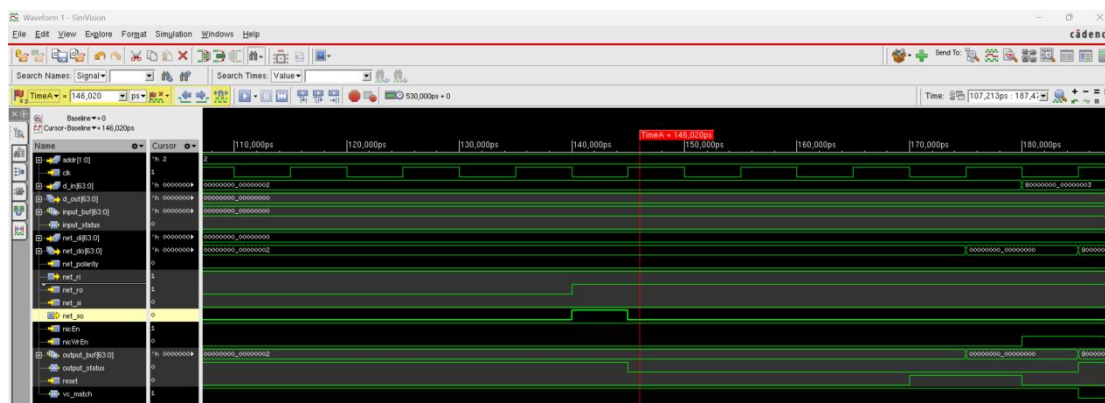
```

===== 2.3.1 WITH BLOCKING (net_ro=0) =====
T=85 ns   rst=1 | nicEn=1 wr=0 addr=10 | out_stat=0 in_stat=0 | net_so=0 net_ro=1 net_ri=1 | VC=0 pol=0
T=95 ns   rst=0 | nicEn=1 wr=1 addr=10 | out_stat=0 in_stat=0 | net_so=0 net_ro=1 net_ri=1 | VC=0 pol=0
T=105 ns  rst=0 | nicEn=1 wr=0 addr=10 | out_stat=1 in_stat=0 | net_so=0 net_ro=0 net_ri=1 | VC=0 pol=0
T=115 ns  rst=0 | nicEn=1 wr=0 addr=10 | out_stat=1 in_stat=0 | net_so=0 net_ro=0 net_ri=1 | VC=0 pol=0
T=125 ns  rst=0 | nicEn=1 wr=0 addr=10 | out_stat=1 in_stat=0 | net_so=0 net_ro=0 net_ri=1 | VC=0 pol=0
T=135 ns  rst=0 | nicEn=1 wr=0 addr=10 | out_stat=1 in_stat=0 | net_so=0 net_ro=0 net_ri=1 | VC=0 pol=0
T=145 ns  rst=0 | nicEn=1 wr=0 addr=10 | out_stat=1 in_stat=0 | net_so=1 net_ro=1 net_ri=1 | VC=0 pol=0
T=155 ns  rst=0 | nicEn=1 wr=0 addr=10 | out_stat=0 in_stat=0 | net_so=0 net_ro=1 net_ri=1 | VC=0 pol=0
T=165 ns  rst=0 | nicEn=1 wr=0 addr=10 | out_stat=0 in_stat=0 | net_so=0 net_ro=1 net_ri=1 | VC=0 pol=0

```



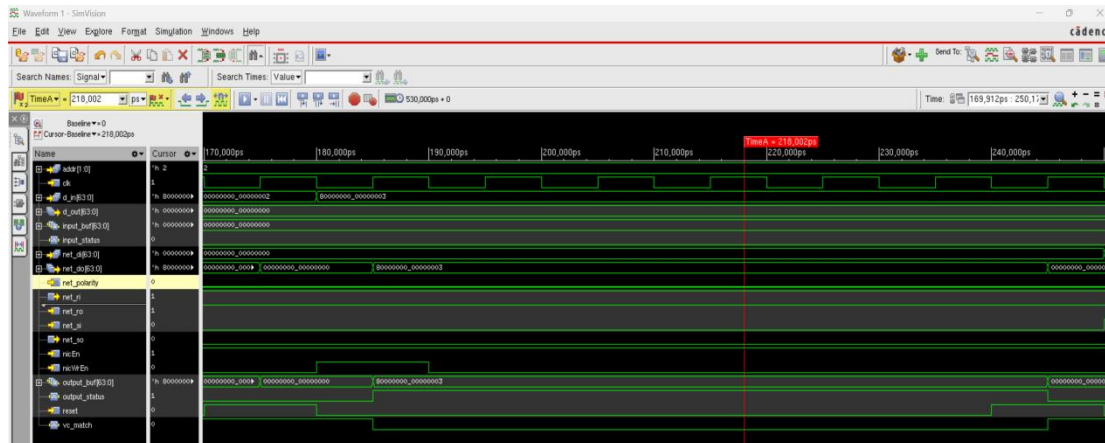
Date being blocked because of **net_ro = 0**.



However, if **net_ro** becomes from 0 to 1, Date could release to router. (**net_so=1**).

2.3.1.2.VC mismatch (NIC to router).

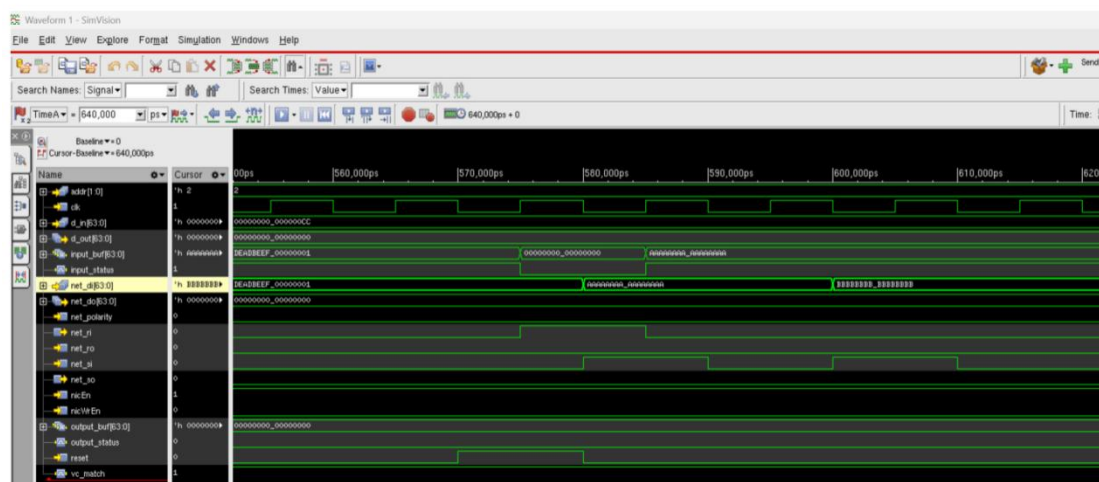
```
==== 2.3.1 WITH BLOCKING (VC mismatch) ====
T=175 ns | rst=1 | nicEn=1 wr=0 addr=10 | out_stat=0 in_stat=0 | net_so=0 net_ro=1 net_ri=1 | VC=0 pol=0
T=185 ns | rst=0 | nicEn=1 wr=1 addr=10 | out_stat=0 in_stat=0 | net_so=0 net_ro=1 net_ri=1 | VC=0 pol=0
T=195 ns | rst=0 | nicEn=1 wr=0 addr=10 | out_stat=1 in_stat=0 | net_so=0 net_ro=1 net_ri=1 | VC=1 pol=0
T=205 ns | rst=0 | nicEn=1 wr=0 addr=10 | out_stat=1 in_stat=0 | net_so=0 net_ro=1 net_ri=1 | VC=1 pol=0
T=215 ns | rst=0 | nicEn=1 wr=0 addr=10 | out_stat=1 in_stat=0 | net_so=0 net_ro=1 net_ri=1 | VC=1 pol=0
T=225 ns | rst=0 | nicEn=1 wr=0 addr=10 | out_stat=1 in_stat=0 | net_so=0 net_ro=1 net_ri=1 | VC=1 pol=0
T=235 ns | rst=0 | nicEn=1 wr=0 addr=10 | out_stat=1 in_stat=0 | net_so=0 net_ro=1 net_ri=1 | VC=1 pol=0
```



`net_so = 0` because of `net_polarity = 0` but MSB of `d_in` is 1. (Mismatching)

2.3.1.3.Router send data to NIC failed (router to NIC).

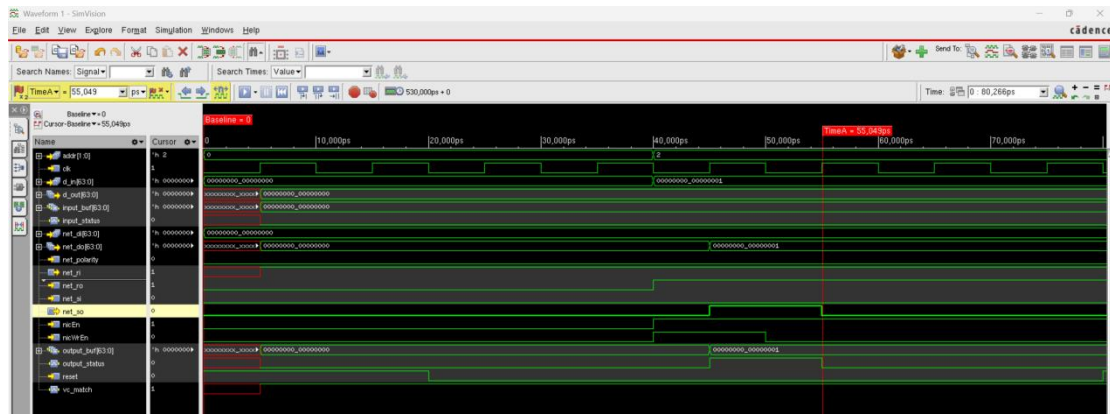
```
==== NEW: ROUTER WRITE FAIL (NIC not ready) ====
T=575 ns | rst=1 | nicEn=1 wr=0 addr=10 | out_stat=0 in_stat=1 | net_so=0 net_ro=0 net_ri=0 | VC=0 pol=0
T=585 ns | rst=0 | nicEn=1 wr=0 addr=10 | out_stat=0 in_stat=0 | net_so=0 net_ro=0 net_ri=1 | VC=0 pol=0
T=595 ns | rst=0 | nicEn=1 wr=0 addr=10 | out_stat=0 in_stat=1 | net_so=0 net_ro=0 net_ri=0 | VC=0 pol=0
T=605 ns | rst=0 | nicEn=1 wr=0 addr=10 | out_stat=0 in_stat=1 | net_so=0 net_ro=0 net_ri=0 | VC=0 pol=0
T=615 ns | rst=0 | nicEn=1 wr=0 addr=10 | out_stat=0 in_stat=1 | net_so=0 net_ro=0 net_ri=0 | VC=0 pol=0
T=625 ns | rst=0 | nicEn=1 wr=0 addr=10 | out_stat=0 in_stat=1 | net_so=0 net_ro=0 net_ri=0 | VC=0 pol=0
T=635 ns | rst=0 | nicEn=1 wr=0 addr=10 | out_stat=0 in_stat=1 | net_so=0 net_ro=0 net_ri=0 | VC=0 pol=0
==== ALL TESTS DONE ====
```



2.3.2.Without Blocking.

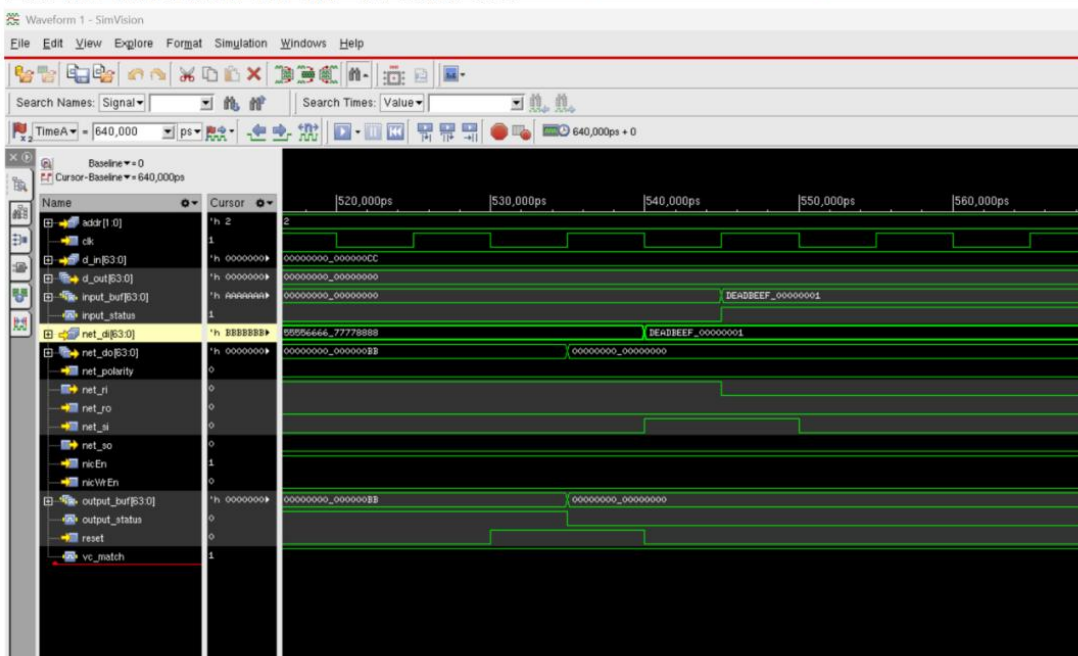
2.3.2.1.NIC to router.

```
==== 2.3.2 WITHOUT BLOCKING ====
T=45 ns | rst=0 | nicEn=1 wr=1 addr=10 | out_stat=0 in_stat=0 | net_so=0 net_ro=1 net_ri=1 | VC=0 pol=0
T=55 ns | rst=0 | nicEn=1 wr=0 addr=10 | out_stat=1 in_stat=0 | net_so=1 net_ro=1 net_ri=1 | VC=0 pol=0
T=65 ns | rst=0 | nicEn=1 wr=0 addr=10 | out_stat=0 in_stat=0 | net_so=0 net_ro=1 net_ri=1 | VC=0 pol=0
T=75 ns | rst=0 | nicEn=1 wr=0 addr=10 | out_stat=0 in_stat=0 | net_so=0 net_ro=1 net_ri=1 | VC=0 pol=0
```

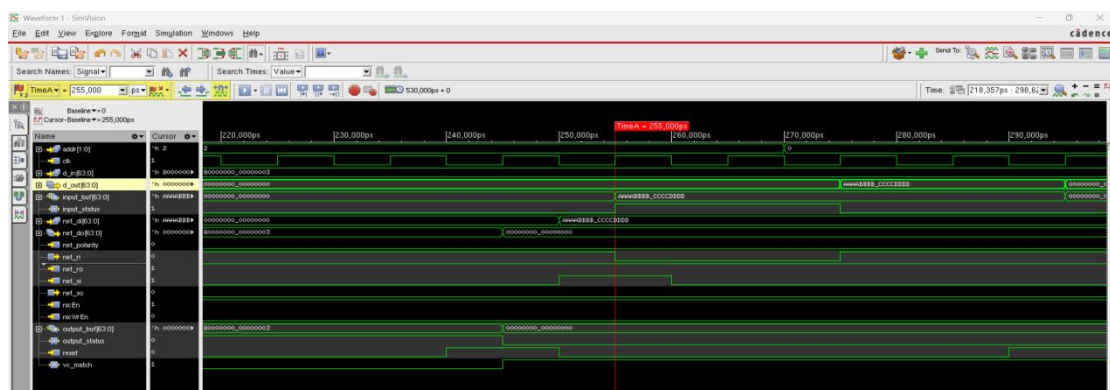
2.3.2.2.Router to NIC.

```
==== NEW: ROUTER WRITE SUCCESS (NIC accepts) ====
T=535 ns | rst=1 | nicEn=1 wr=0 addr=10 | out_stat=1 in_stat=0 | net_so=0 net_ro=0 net_ri=1 | VC=0 pol=0
T=545 ns | rst=0 | nicEn=1 wr=0 addr=10 | out_stat=0 in_stat=0 | net_so=0 net_ro=0 net_ri=1 | VC=0 pol=0
T=555 ns | rst=0 | nicEn=1 wr=0 addr=10 | out_stat=0 in_stat=1 | net_so=0 net_ro=0 net_ri=0 | VC=0 pol=0
T=565 ns | rst=0 | nicEn=1 wr=0 addr=10 | out_stat=0 in_stat=1 | net_so=0 net_ro=0 net_ri=0 | VC=0 pol=0
==== NEW: ROUTER WRITE FAIL (NIC not ready) ====
```



2.4.Waveform showing interactions between processor and NIC

2.4.1.Load operation with buffer being available.



```

===== 2.4.1 LOAD AVAILABLE =====
T=245 ns | rst=1 | nicEn=1 wr=0 addr=10 | out_stat=1 in_stat=0 | net_so=0 net_ro=1 net_ri=1 | VC=1 pol=0
T=255 ns | rst=0 | nicEn=1 wr=0 addr=10 | out_stat=0 in_stat=0 | net_so=0 net_ro=1 net_ri=1 | VC=0 pol=0
T=265 ns | rst=0 | nicEn=1 wr=0 addr=10 | out_stat=0 in_stat=1 | net_so=0 net_ro=1 net_ri=0 | VC=0 pol=0
T=275 ns | rst=0 | nicEn=1 wr=0 addr=00 | out_stat=0 in_stat=1 | net_so=0 net_ro=1 net_ri=0 | VC=0 pol=0
T=285 ns | rst=0 | nicEn=1 wr=0 addr=00 | out_stat=0 in_stat=0 | net_so=0 net_ro=1 net_ri=1 | VC=0 pol=0

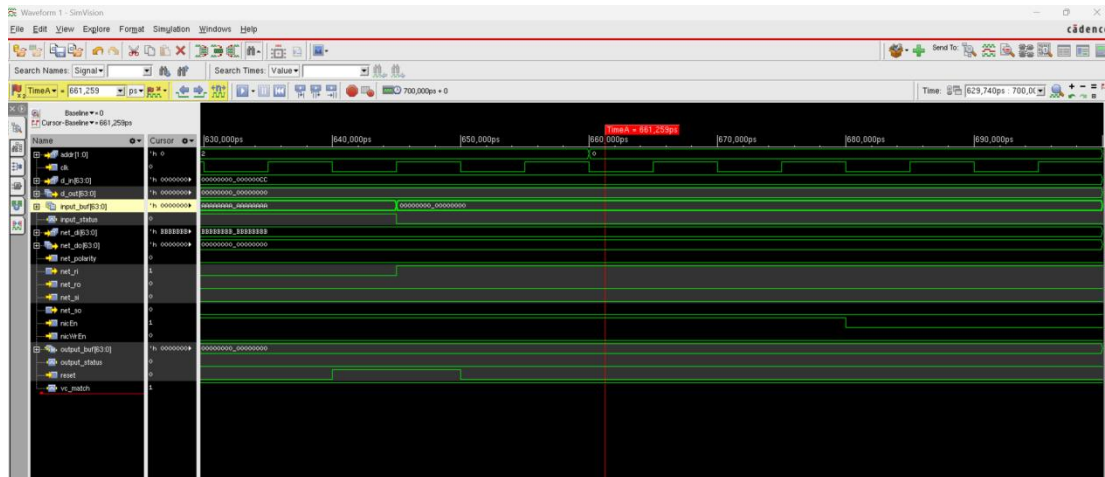
```

2.4.2. Load operation with buffer being unavailable.

```

===== 2.4.2 LOAD UNAVAILABLE (BUFFER EMPTY) =====
T=645 ns | rst=1 | nicEn=1 wr=0 addr=10 | out_stat=0 in_stat=1 | net_so=0 net_ro=0 net_ri=0 | VC=0 pol=0
T=655 ns | rst=0 | nicEn=1 wr=0 addr=10 | out_stat=0 in_stat=0 | net_so=0 net_ro=0 net_ri=1 | VC=0 pol=0
T=665 ns | rst=0 | nicEn=1 wr=0 addr=00 | out_stat=0 in_stat=0 | net_so=0 net_ro=0 net_ri=1 | VC=0 pol=0
T=675 ns | rst=0 | nicEn=1 wr=0 addr=00 | out_stat=0 in_stat=0 | net_so=0 net_ro=0 net_ri=1 | VC=0 pol=0
T=685 ns | rst=0 | nicEn=0 wr=0 addr=00 | out_stat=0 in_stat=0 | net_so=0 net_ro=0 net_ri=1 | VC=0 pol=0
T=695 ns | rst=0 | nicEn=0 wr=0 addr=10 | out_stat=0 in_stat=0 | net_so=0 net_ro=0 net_ri=1 | VC=0 pol=0
===== ALL TESTS DONE =====

```

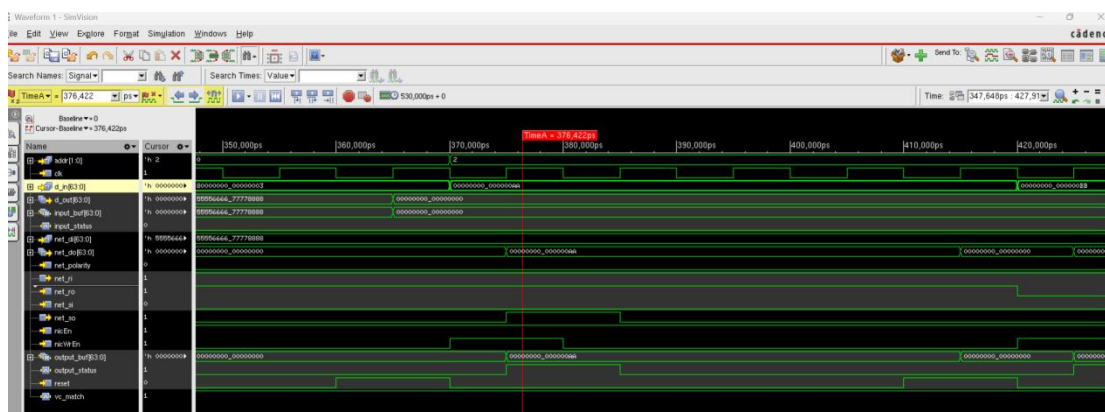


2.4.3. Store operations with buffer being available

```

===== 2.4.3 STORE AVAILABLE =====
T=365 ns | rst=1 | nicEn=1 wr=0 addr=00 | out_stat=0 in_stat=0 | net_so=0 net_ro=1 net_ri=1 | VC=0 pol=0
T=375 ns | rst=0 | nicEn=1 wr=1 addr=10 | out_stat=0 in_stat=0 | net_so=0 net_ro=1 net_ri=1 | VC=0 pol=0
T=385 ns | rst=0 | nicEn=1 wr=0 addr=10 | out_stat=1 in_stat=0 | net_so=1 net_ro=1 net_ri=1 | VC=0 pol=0
T=395 ns | rst=0 | nicEn=1 wr=0 addr=10 | out_stat=0 in_stat=0 | net_so=0 net_ro=1 net_ri=1 | VC=0 pol=0
T=405 ns | rst=0 | nicEn=1 wr=0 addr=10 | out_stat=0 in_stat=0 | net_so=0 net_ro=1 net_ri=1 | VC=0 pol=0

```



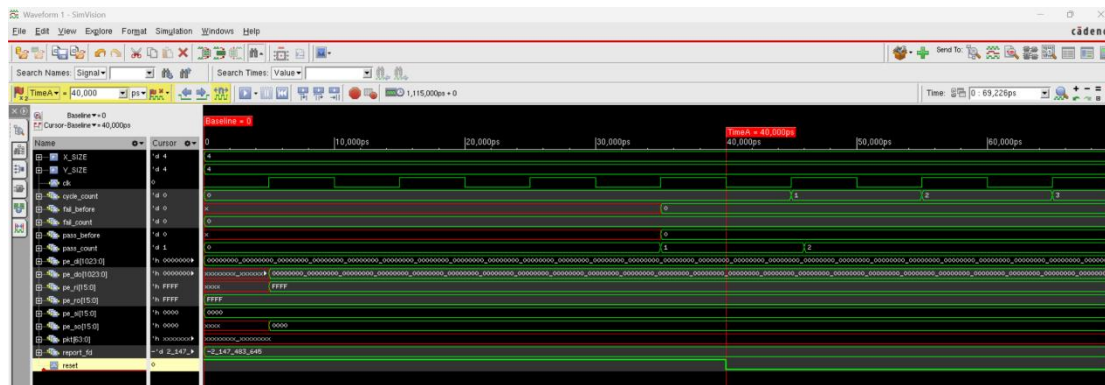
2.4.4. Store operations with buffer being unavailable.

```

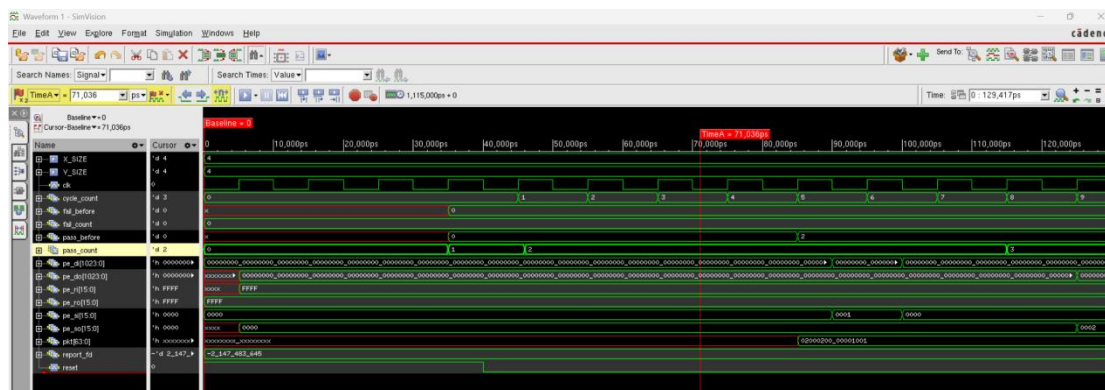
===== 2.4.4 STORE UNAVAILABLE =====
T=415 ns | rst=1 | nicEn=1 wr=0 addr=10 | out_stat=0 in_stat=0 | net_so=0 net_ro=1 net_ri=1 | VC=0 pol=0
T=425 ns | rst=0 | nicEn=1 wr=1 addr=10 | out_stat=0 in_stat=0 | net_so=0 net_ro=0 net_ri=1 | VC=0 pol=0
T=435 ns | rst=0 | nicEn=1 wr=0 addr=10 | out_stat=1 in_stat=0 | net_so=0 net_ro=0 net_ri=1 | VC=0 pol=0
T=445 ns | rst=0 | nicEn=1 wr=0 addr=10 | out_stat=1 in_stat=0 | net_so=0 net_ro=0 net_ri=1 | VC=0 pol=0
T=455 ns | rst=0 | nicEn=1 wr=0 addr=10 | out_stat=1 in_stat=0 | net_so=0 net_ro=0 net_ri=1 | VC=0 pol=0
T=465 ns | rst=0 | nicEn=1 wr=0 addr=10 | out_stat=1 in_stat=0 | net_so=0 net_ro=0 net_ri=1 | VC=0 pol=0
T=475 ns | rst=0 | nicEn=1 wr=0 addr=10 | out_stat=1 in_stat=0 | net_so=0 net_ro=0 net_ri=1 | VC=0 pol=0
T=485 ns | rst=0 | nicEn=1 wr=0 addr=10 | out_stat=1 in_stat=0 | net_so=0 net_ro=0 net_ri=1 | VC=0 pol=0
T=495 ns | rst=0 | nicEn=1 wr=0 addr=10 | out_stat=1 in_stat=0 | net_so=0 net_ro=0 net_ri=1 | VC=0 pol=0
T=505 ns | rst=0 | nicEn=1 wr=0 addr=10 | out_stat=1 in_stat=0 | net_so=0 net_ro=0 net_ri=1 | VC=0 pol=0
T=515 ns | rst=0 | nicEn=1 wr=0 addr=10 | out_stat=1 in_stat=0 | net_so=0 net_ro=0 net_ri=1 | VC=0 pol=0
T=525 ns | rst=0 | nicEn=1 wr=0 addr=10 | out_stat=1 in_stat=0 | net_so=0 net_ro=0 net_ri=1 | VC=0 pol=0
===== ALL TESTS DONE =====

```

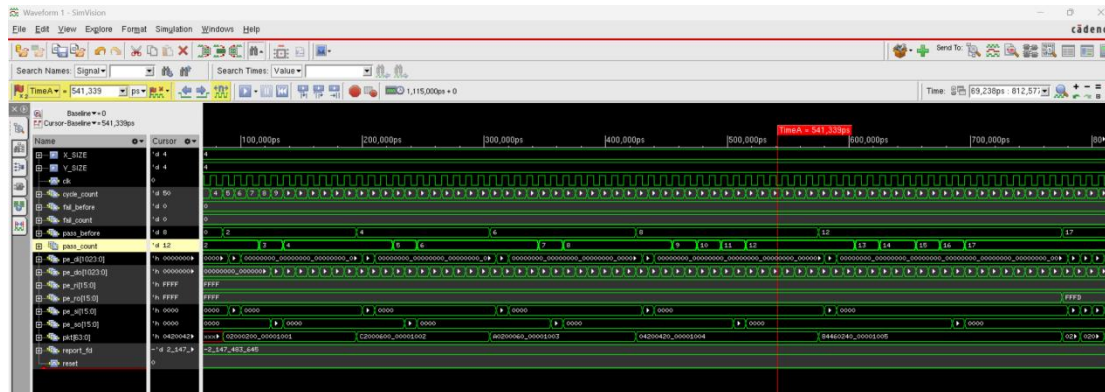

3.1. One waveform showing reset signal operation.



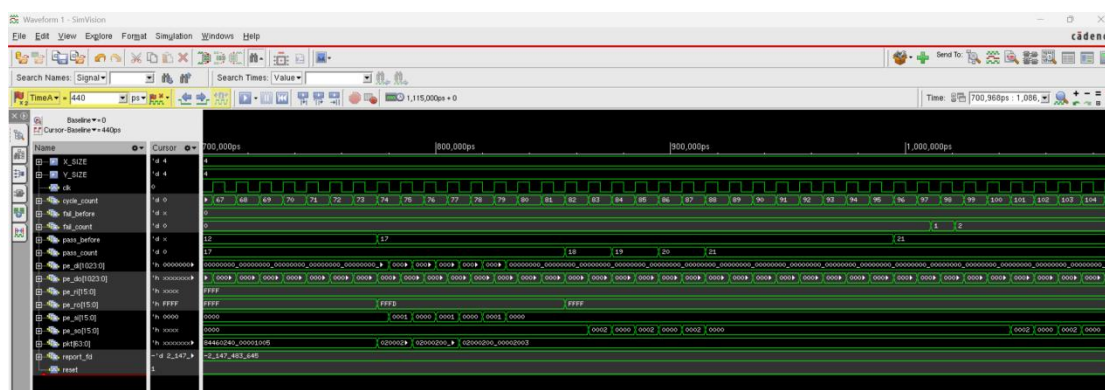
3.2. Correct hop value updates.



3.3. Six test cases covering different routing and switching conditions

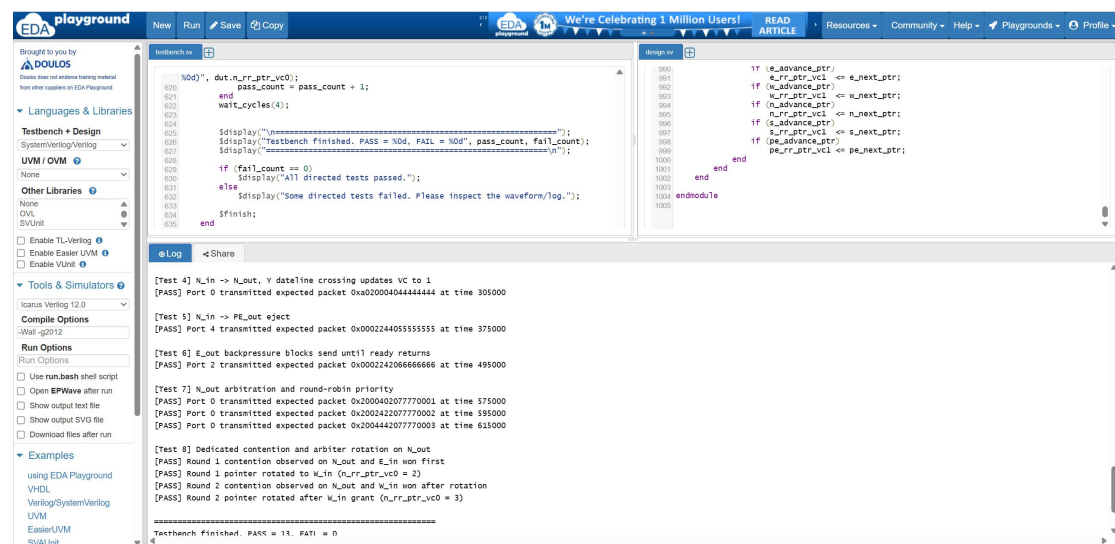


3.4. Waveforms showing router-to-router handshaking, both with and without blocking (Refer to Figures 4–5 in the project PDF)



3.5. At least one test case showing arbiter behavior when contention occurs, clearly .showing priority rotation

```
tb_torus_4x4_network report
1 tb_torus_4x4_network report
2 =====
3 TEST 1 - reset behavior : PASS (delta pass=1, delta fail=0)
4 TEST 2 - single-hop East, non-blocking handshake : PASS (delta pass=2, delta fail=0)
5 TEST 3 - single-hop West wrap-around : PASS (delta pass=2, delta fail=0)
6 TEST 4 - single-hop North wrap-around : PASS (delta pass=2, delta fail=0)
7 TEST 5 - multi-hop X then Y with hop update checks : PASS (delta pass=4, delta fail=0)
8 TEST 6 - X dateline crossing followed by X1 to Y0 phase switch : PASS (delta pass=5, delta fail=0)
9 TEST 7 - blocking handshake on a router-to-router link : PASS (delta pass=4, delta fail=0)
10 TEST 8 - deterministic contention and arbiter priority rotation at router (1,1) N_out : FAIL (delta pass=0, delta fail=2)
11
12 TB DONE: pass_count=21 fail_count=2
13 =====
```



The screenshot shows the EDA Playground interface. The top section displays a Verilog testbench for a torus network. The testbench includes a report header, a list of tests (TEST 1 to TEST 8), and a summary line 'TB DONE: pass_count=21 fail_count=2'. The test results show that TEST 8 failed due to deterministic contention and arbiter priority rotation at router (1,1) N_out. The bottom section shows the Verilog code for the testbench, which includes a module 'tb_torus_4x4_network' and a testbench 'tb_torus_4x4_network_testbench'. The testbench code includes a 'report' function that prints the test results. The execution results show the testbench running successfully, with all tests passing except for TEST 8, which failed. The testbench output shows the test results for each test, including the pass/fail status and the delta pass/fail values. The testbench also shows the test results for the entire testbench, including the total pass/fail count.